

Delivery Failure Lab

SLA reliability and sales exposure

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DataCo supply chain study



Delivery reliability needs an operational pilot

57.31%

of eligible orders missed their scheduled SLA

100.00% / 79.99%

Late rates for First Class and Second Class

14,995 late orders

Standard Class accounts for \$7.52M in associated net sales

10% relative improvement would mean 3,605 fewer late orders

Project scope and objective

DECISION TO SUPPORT

Which shipping services should the team investigate first?

Historical coverage

Jan 2015–Jan 2018
65,752 unique orders
Five markets

Analysis scope

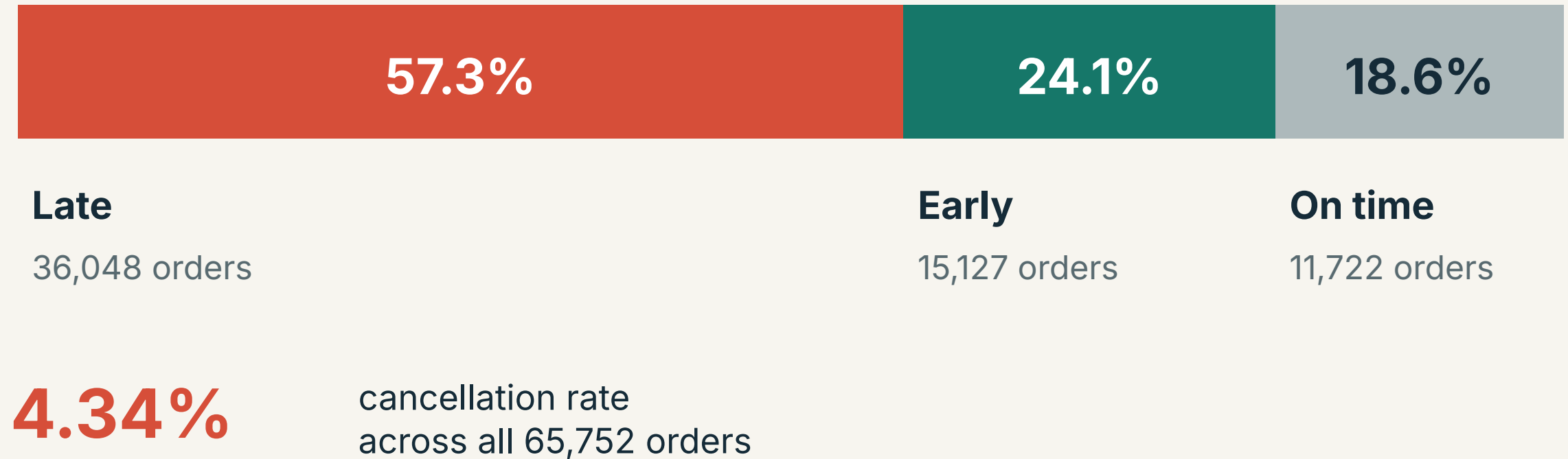
Delivery SLA, shipping mode, delay severity, sales exposure and improvement scenarios

One order receives one vote in delivery KPIs

Processing stage	Count	Meaning
Raw dataset	180,519	Order-item rows
Order-level dataset	65,752	One row per unique Order Id
Canceled shipping status	2,855	Excluded from the SLA denominator
Eligible orders	62,897	Population for delivery reliability

65,752 unique orders form the analysis base

More than half of eligible orders missed their scheduled SLA



First Class and Second Class have the highest late rates

First Class

9,602 eligible orders



100.00%

Second Class

12,256 eligible orders



79.99%

Same Day

3,407 eligible orders



48.37%

Standard Class

37,632 eligible orders



39.85%

Validate SLA definitions and feasibility for both services

Late status uses a different cutoff for each shipping mode

Shipping mode	SLA	Recorded days	Late when	Late rate
First Class	1 day	2 days	>1 day	100.00%
Second Class	2 days	2–6 days	3–6 days	79.99%
Standard Class	4 days	2–6 days	5–6 days	39.85%
Same Day	0 days	0–1 calendar day	1 calendar day	48.37%

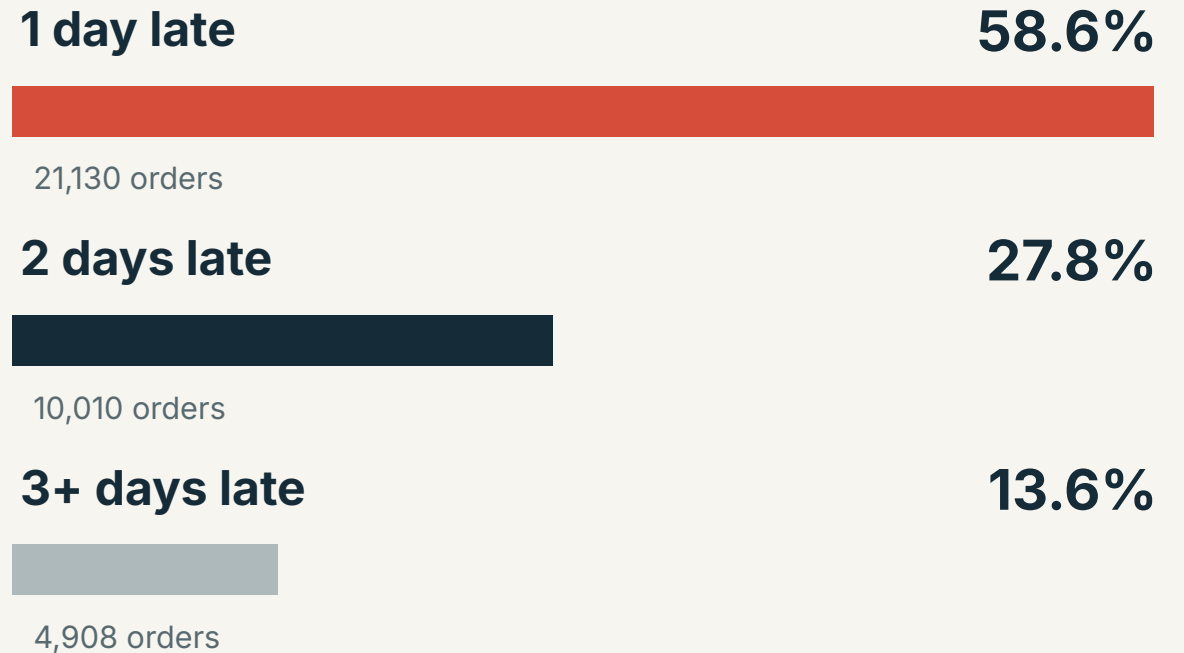
The same 2–6 day range produces very different late rates because the SLA cutoffs differ

Second Class is late at 3–6 days, while Standard Class is late only at 5–6 days. The dataset does not explain the operational cause of these durations.

One-day delays account for most late orders

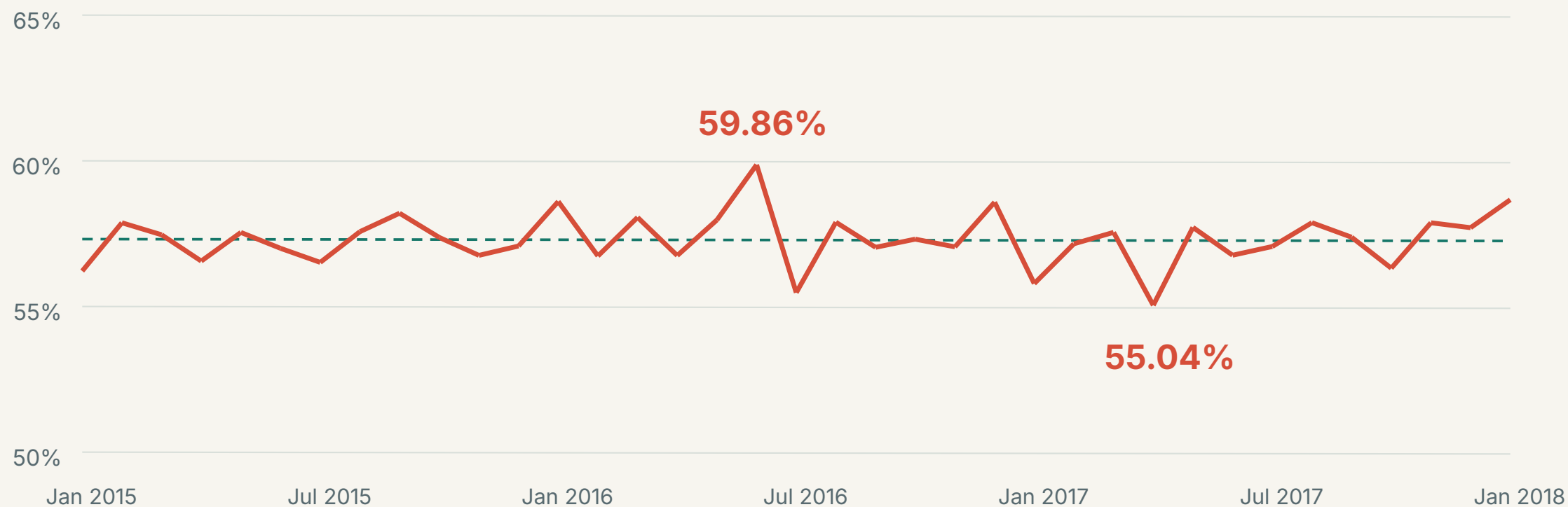
58.6%

21,130 orders missed their SLA by one day



Split the one-day group by shipping mode before choosing a process test

Aggregate monthly late rate stays around 57%



37 monthly observations

Full-period baseline 57.31%

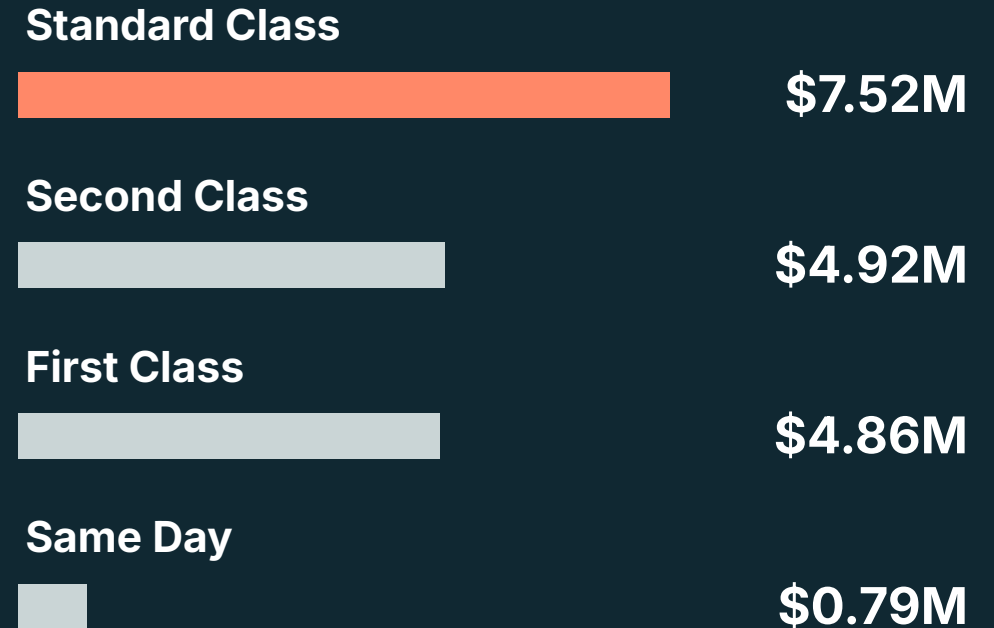
Monthly range: 55.04%–59.86%. The rate axis starts at 50%. January 2018 includes Pacific Asia only.

Net sales associated with late orders

\$18.08M

Standard Class represents
41.6% of this exposure

**Recorded profit on late orders: \$2.14M.
This is not an estimate of profit lost.**



Rate and volume support different intervention priorities

Shipping mode	Eligible orders	Late orders	Late rate	Sales exposure
First Class	9,602	9,602	100.00%	\$4.86M
Second Class	12,256	9,803	79.99%	\$4.92M
Standard Class	37,632	14,995	39.85%	\$7.52M
Same Day	3,407	1,648	48.37%	\$0.79M

First Class and Second Class

Validate SLA definitions and feasibility.

Standard Class

Prioritize further investigation by late volume.

**A 10% relative improvement
would lower the late rate to 51.58%**

3,605

fewer late orders

32,443 remaining late orders

Historical baseline

57.31%

10% improvement scenario

51.58%

5.73 percentage points lower

Six months, starting with a 90-day pilot

Phase	Focus	Proposed owner	Decision gate
Days 1–30	Validate SLA rules, timestamps and process data	Operations + analyst	Reliable baseline and feasible pilot
Days 31–60	Test a change selected from the initial investigation	Logistics + warehouse	Lower late rate with stable cost
Days 61–90	Compare with a comparable control group or baseline	Analyst + finance	Scale, revise or stop
Days 91–180	Scale if the improvement and cost controls hold	Operations lead	10% relative reduction target

Proposed owners and timing depend on operational access and agreement on the pilot.

Delivery gains need cost and customer guardrails

Metric	Historical baseline	Pilot success criterion
Late delivery rate	57.31% of eligible orders	At least 10% relative reduction
Cancellation rate	4.34% of all orders	No adverse change from pilot baseline
Shipping cost per order	Additional data required	Within an agreed cost threshold
Delay complaints / claims	Additional data required	Stable or improving after the change

Compare the same service mix and eligibility rules

A live pilot requires a current, comparable baseline. Historical values illustrate the proposed target.

The next decision: an operational pilot

Review SLAs First & Second Class

Validate the scheduled promise
and shipment timestamp definitions.

Explore the Tableau dashboard

Investigate Standard Class

The service with the
largest number of late orders.

Metric definitions

Metric	Definition
Order grain	One row per Order Id; monetary fields sum across order items
Eligible order	Delivery_Status differs from Shipping canceled
Late order	Eligible order with Actual_Shipping_Days > Scheduled_Shipping_Days
Late delivery rate	Late orders divided by eligible orders
Cancellation rate	Shipping-canceled orders divided by all orders
Late sales exposure	Sum of Net_Sales for late eligible orders

Version 2 makes the analytical basis explicit

Area	Version 1 presentation	Version 2 approach
Delivery population	Some volume comparisons used item rows	One order per Order Id. Cancellations excluded only from SLA outcome calculations
Financial language	Profit at risk carried ambiguous loss language	Sales and profit exposure with explicit meaning
Causal interpretation	Shipping mode described as a root cause	Association partly reflects SLA rules. Operational causes remain untested
Improvement target	10% objective without a consistent scenario basis	Relative target with explicit FINAL scenario formulas

Interpret the shipping-mode link alongside the SLA rules

Shipping mode		0.481
Order region		0.012
Order year		0.001
Market		0.000
Customer segment		0.000
Order complexity		0.000

0.481

Each mode determines its own SLA cutoff. Part of this association reflects the definition of late status.

Same Day depends on the calendar boundary

12 hours

between recorded order and shipping timestamps for all 3,407 eligible Same Day orders

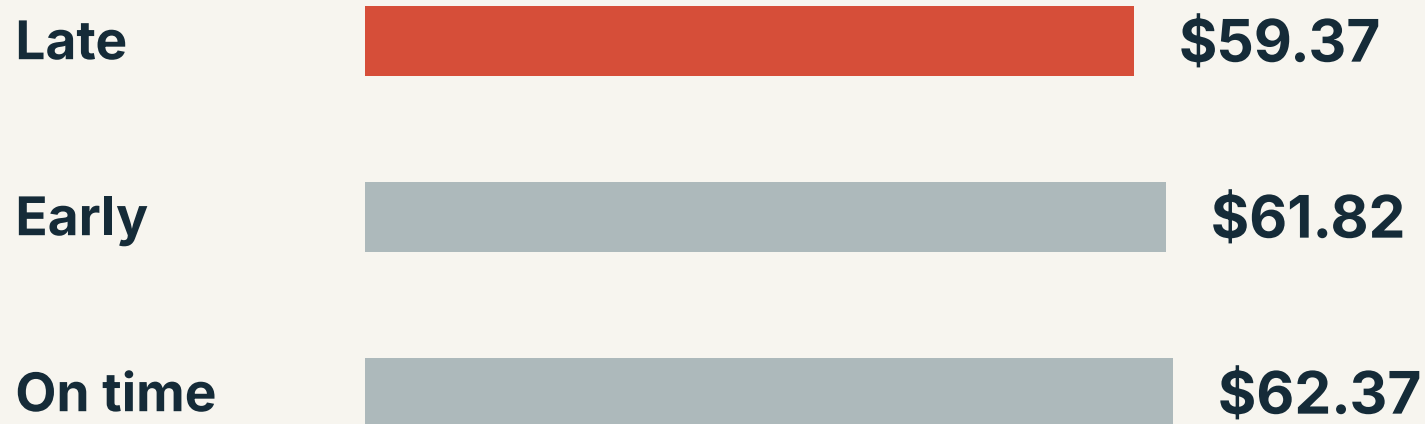
Calendar-date outcome	Eligible orders	Recorded actual days	SLA outcome
Same date	1,759	0	On time
Next date	1,648	1	Late

The calendar boundary changes the late flag

Validate service cutoff rules and the meaning of each timestamp.

Same elapsed duration, different calendar dates. The zero-day SLA uses the recorded calendar-day convention.

Average recorded profit ranges from \$59.37 to \$62.37 per order



\$2.70

lower average profit
for late orders versus
all non-late orders

Economic harm requires returns, compensation, cost and churn data

Scenario reference: 0%–25% relative improvement

Relative target	Avoided orders	Remaining late	Projected rate	Exposure reduction
0%	0	36,048	57.31%	\$0.00M
5%	1,802	34,246	54.45%	\$0.90M
10%	3,605	32,443	51.58%	\$1.81M
15%	5,407	30,641	48.72%	\$2.71M
20%	7,210	28,838	45.85%	\$3.62M
25%	9,012	27,036	42.98%	\$4.52M

Projected late orders = ROUND(baseline late orders × (1 – target))

Fixed eligible volume. Exposure reduction = baseline late sales × target. Scenario values only.

Sources and interpretation limits

Analysis assets

DataCoSupplyChainDataset.csv

01_data_quality_audit_COMPLETE.ipynb

Tableau FINAL workbook

Interpretation limits

Historical, observational data

January 2018 covers Pacific Asia only

Incomplete process milestones and cost data

Scenario assumptions require validation

Public project dashboard